

Venkata Siva Naga Sai Aravind Kollipara

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SUMMARY

Software Engineer with 5+ years of experience in **scalable back end systems**, **cloud infrastructure**, and **data optimization** using **Java**, **Python**, **AWS** and **OCI**. Recently focused on AI-driven applications, including **LLM-based assistants**, **RAG** with **ChromaDB**, and multi-modal tools using **LangGraph**, **Cohere**, and **Gemini**. Skilled in **CI/CD**, **TDD**, and serverless architectures, with a track record of building robust, intelligent solutions and user-friendly interfaces in fast-paced Agile environments.

EXPERIENCE

Senior Member of Technical Staff, Oracle | Austin, United States of America

11/2025 – Present

- Contributed to the back-end architecture of **OCI's Zero Trust Packet Routing (ZPR)**, a distributed network-layer enforcement system that evaluates packet flows across compute and database services at scale.
- Validated attribute-driven access control logic for service-service communication within **Virtual Cloud Networks (VCNs)**, improving isolation and reducing lateral movement risks in multi-tenant environments
- Led **region build** and service onboarding to ZPR, ensuring consistent policy enforcement across environments.
- Improved secure internal communication between compute instances, databases, and private endpoints through policies.
- Troubleshoot and resolved policy enforcement issues during service releases and strengthening internal service security.

Software Development Engineer, Amazon | Seattle, United States of America

10/2022 – 11/2025

- Designed and implemented scalable cloud-based backend scheduling services using **Java**, while leveraging **CI/CD** pipelines.
- Spearheaded the scaling of the back-end scheduling service, improving the system capacity from **500** to **12000** agents. Optimized **API** response time by **50%** and ensured **99.99%** system uptime through performance tuning and system monitoring.
- Implemented a **back-filling** mechanism to restore missing shift data in **data lake** by reprocessing affected records, ensuring data integrity and preventing future data loss in the scheduling system.
- Led the development of the **Shift Exchange feature**, allowing agents to **swap shifts dynamically**, improving operational flexibility and reducing manual scheduling interventions by **85%**.
- Engineering and optimized high-performance **REST API's**, achieving a **40%** reduction in latency through efficient endpoint design and implementation of **React**, **TypeScript** front-end architecture with in a microservices ecosystems.
- Optimized **DynamoDB** and **RDS** by implementing efficient indexing strategies (GSI and LSCI), and optimizing partition keys.
- Drove migration of **Java Development Kit (JDK)** from **JDK8** to **JDK17** and upgraded Cloud Deployment Kit (**CDK**) from **CDK V1** to **CDK V2**, ensuring compatibility and utilizing new features to enhance infrastructure-as-code practices.
- Collaborated with cross-functional teams to prioritize and implement Santander-specific requirements.

Application Development Associate, Accenture | Hyderabad, India

01/2021 – 06/2021

- Developed a hotel recommendation system** leveraging **C#**, **.NET**, and **GraphQL on Microsoft Azure**, integrating **ratings**, **awards**, **proximity**, and **user preferences**, which **improved recommendation accuracy by 25%**.
- Implemented real-time data updates and personalized features, collaborating with clients to meet requirements while developing executing test cases to improve test quality and ensure code reliability.
- Achieved a **30%** increase in user satisfaction and a **20%** rise in booking rates through optimized functionality.

Database Developer, BNP Paribas | Chennai, India

01/2020 – 12/2020

- Resolved data processing issues, optimized **SQL** queries and database schema for better performance and scalability which resulted in improving system uptime by **9%**.
- Tested and developed scalable back-end solutions to efficiently support a larger number of agents.
- Created **Tableau** dashboards to visualize sales and compensation data, boosting reporting efficiency by **20%**.

SKILLS

Programming Languages: Python, Java, JavaScript, Golang, C++, .NET, HTML, CSS, C#

Databases: SQL Server, MySQL, PostgreSQL, Snowflake, Databricks, Redshift, MongoDB, CosmosDB

Frameworks and Tools: ReactJS, Django, Solr, AWS, GCP, RESTful API, JIRA, Git, Kafka, Kubernetes, Spring Boot, Linux, GraphQL, CI/CD, Agent AI, Cohere, LLM, ChromaDB, Ngrok, LangSmith, Gorq, Chatbot, Agents, GEMINI, LLaMA, OpenAI

KEY PROJECTS

AI Assistant Hub on Ngrok – Multi-modal ChatBot with Image Generation

- Built a dual-mode AI assistant using **LangGraph**, **Groq LLaMA 3**, and **Google Gemini**, enabling real-time text-based conversation and prompt-to-image generation with tool-based reasoning via Tavily Search.
- Designed a **Streamlit** interface with **ChromaDB-based RAG** for PDF Q&A, multimodal output, and **Ngrok** for public domain.

Neural Networks for Bird Species Classification

- Worked on a **Deep Learning** project using convolutional Neural Networks Methodology for bird species identification, achieving **90% accuracy**. Preprocessed audio data through Clustering, Re-Sampling, segment identification, and spectrogram computation.
- Leveraged dropout layers in the model design to prevent over-fitting, enhanced model performance to **93%**.

EDUCATION

Master of Science in Computer Science, University at Buffalo, NY.

08/2021 – 08/2022

Bachelor of Technology in Information Technology, SRMIST, India.

08/2016 – 05/2020